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# SURE A EXAMINATION COUNCIL

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**UCE CHEMISTRY 2 2026**
**MARKING GUIDE**

## CHEMISTRY PRACTICAL

River-water alkalinity investigation — worked model answer

**Important note:** The uploaded scan could not be read reliably as text. This printable document organises the worked model answer already prepared in the conversation. The numerical titration readings are **illustrative practice data**, not measurements extracted from the examination paper. Replace them with actual readings if the paper supplies data.

### 1. Title, aim and hypothesis

**Title:** Determination of the variation in river-water alkalinity with distance from a suspected pollution source.

**Aim:** To investigate how the neutralisable alkalinity of river-water samples varies with distance from the pollution source, using dilute hydrochloric acid.

**Hypothesis:** The neutralisable alkalinity of the river water decreases as distance from the pollution source increases.

## 2. Variables

| Variable | Description |
|---|---|
| Independent | Distance of the sampling point from the pollution source. |
| Dependent | Volume of dilute hydrochloric acid required to reach the indicator end-point. |
| Controlled | Volume of water sample, acid concentration, indicator type and amou and titration procedure. |

## 3. Apparatus and materials

**Apparatus:** Burette, pipette (25 $cm^3$), conical flask, retort stand and clamp, funnel, wash bottle, dropper, white tile and measuring cylinder.

**Materials:** Water samples BA1, BA2 and BA3; dilute hydrochloric acid (BA4); phenolphthalein indicator; distilled water.

## 4. Risks and precautions

| Possible risk | Precaution |
|---|---|
| Hydrochloric acid may irritate skin. | Wear suitable gloves and avoid direct contact. |
| Chemical splashes may affect the eyes. | Wear safety goggles and handle the burette carefully. |
| Glassware may break. | Handle apparatus carefully; report and safely clear breakages. |
| Water samples may be contaminated. | Do not taste any sample; wash hands after the experiment. |
| Chemical spills may occur. | Inform the supervisor and clean spills using the approved laboratory procedure. |

## 5. Experimental procedure

1. The burette was cleaned and rinsed with dilute hydrochloric acid.
2. The burette was filled with the acid, ensuring that the jet contained no air bubbles; the initial reading was recorded.
3. Using a pipette, 25 $cm^3$ of sample BA1 was transferred into a clean conical flask.
4. Two drops of phenolphthalein indicator were added and the flask was placed on a white tile.
5. Hydrochloric acid was run into the flask while the flask was gently swirled.
6. As the end-point approached, acid was added dropwise until the pink colour just disappeared.
7. The final burette reading was recorded and the titre was calculated.
8. The titration was repeated until six readings had been obtained for BA1.
9. The same procedure was repeated for BA2 and BA3, using clean apparatus as appropriate.

## 6. Results and observations

The table below uses illustrative model readings for practice. The volume of each water sample was 25 $cm^3$.

| Reading | BA1 ($cm^3$) | BA2 ($cm^3$) | BA3 ($cm^3$) |
|---|---|---|---|
| 1 | 18.4 | 12.8 | 5.0 |
| 2 | 18.5 | 12.9 | 5.1 |
| 3 | 18.6 | 13.0 | 5.2 |
| 4 | 18.7 | 13.1 | 5.3 |
| 5 | 18.5 | 13.0 | 5.2 |
| 6 | 18.6 | 13.2 | 5.1 |
| Mean | 18.55 | 13.00 | 5.15 |

**Mean titre calculation for BA1:**

(18.4 + 18.5 + 18.6 + 18.7 + 18.5 + 18.6) ÷ 6 = 111.3 ÷ 6 = **18.55 $cm^3$**.

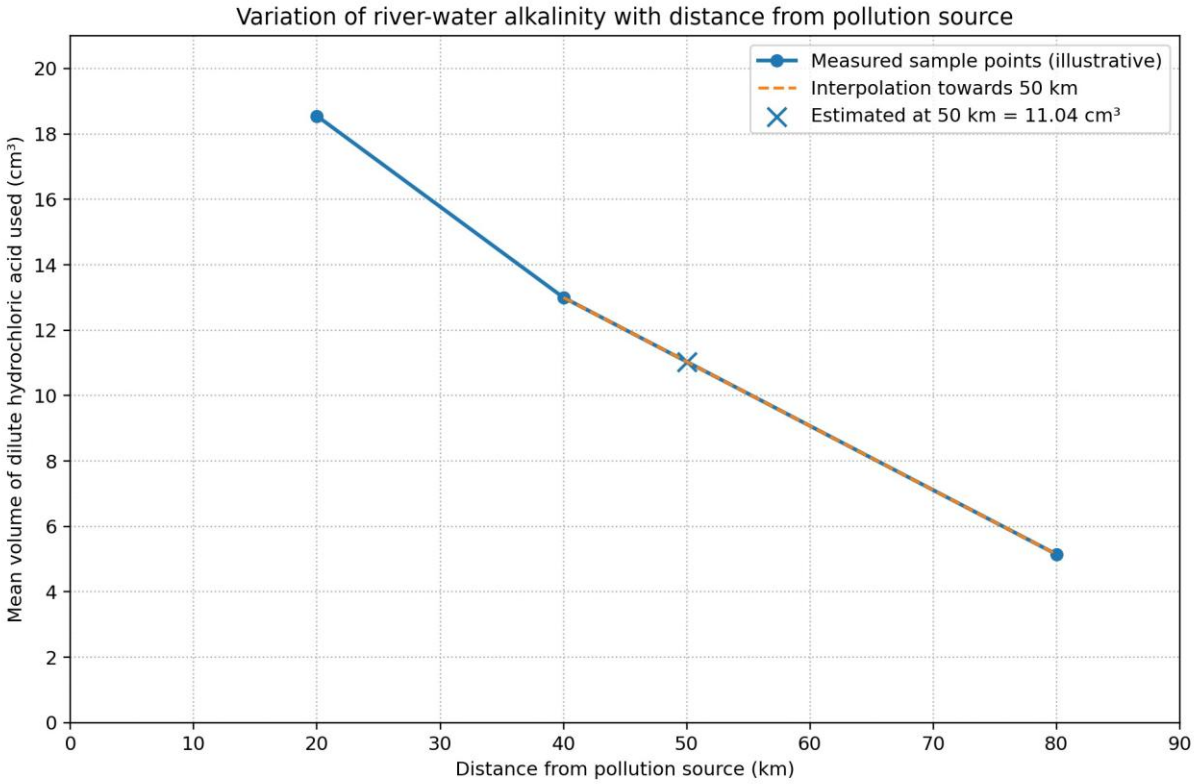
**Mean titre calculation for BA2:** 78.0 ÷ 6 = **13.00 $cm^3$**.

**Mean titre calculation for BA3:** 30.9 ÷ 6 = **5.15 $cm^3$**.

| Sample | Qualitative observation |
|---|---|
| BA1 | The largest volume of acid was required to reach the end-point. |
| BA2 | A smaller volume of acid was required than for BA1. |
| BA3 | The smallest volume of acid was required. |

## 7. Graphical presentation

Plot distance from the pollution source on the horizontal (x) axis and mean volume of hydrochloric acid used on the vertical (y) axis. Label both axes with units and use a clear, even scale.




The graph is based on illustrative sample positions of 20 km, 40 km and 80 km, paired with the model mean titres for BA1, BA2 and BA3 respectively.

## 8. Estimation at the inaccessible point (50 km)

Using linear interpolation between 40 km (13.00 cm³) and 80 km

(5.15 cm³): $y = 13.00 + [(50 - 40) \div (80 - 40)] \times (5.15 - 13.00)$

$y = 13.00 + 0.25 \times (-7.85) = 13.00 - 1.9625$

**Estimated mean titre at 50 km = 11.04 cm³ (approximately).**

This is an estimate from the illustrative graph, not a measured value.

## 9. Interpretation and deduction

- At 20 km, the illustrative mean titre was highest (18.55 cm³), suggesting the greatest neutralisable alkalinity amongthe three samples.
- At 40 km, the mean titre decreased to 13.00 cm³, suggesting lower neutralisable alkalinity.
- At 80 km, the mean titre was lowest (5.15 cm³), suggesting the lowest neutralisable alkalinity among the samples.
- The pattern suggests a negative relationship between distance from the suspected source and neutralisablealkalinity.

**Deduction:** The illustrative results suggest that neutralisable alkalinity decreases with increasing distance from the pollution source. Dilution and mixing with river water of lower alkalinity may contribute to this pattern.

## 10. Conclusion

The illustrative model results support the hypothesis that neutralisable alkalinity decreases with increasing distance from the pollution source. The highest mean titre was recorded for BA1 and the lowest for BA3.

## 11. Recommendations to district authorities

- Identify and control the source of the suspected alkaline discharge.
- Require industries to treat wastewater before releasing it into the environment.
- Monitor river-water quality regularly at several points upstream and downstream.
- Investigate the inaccessible 50 km point when safe access becomes possible.
- Enforce environmental regulations and educate nearby communities about preventing water pollution.
- Use additional water-quality tests to assess pH and other possible contaminants.

## 12. Scientific limitation

The volume of acid required in this titration gives an indication of the sample's neutralisable alkalinity under the stated conditions. It does not, by itself, identify the pollutant, establish the exact pH, or determine whether the water is safe for drinking. Further tests and verified field measurements are needed.

**Exam reminder:** In the examination, use the actual sample labels, distances, acid concentration, indicator and readings supplied in the question. Do not copy the illustrative readings above if the paper provides different data.